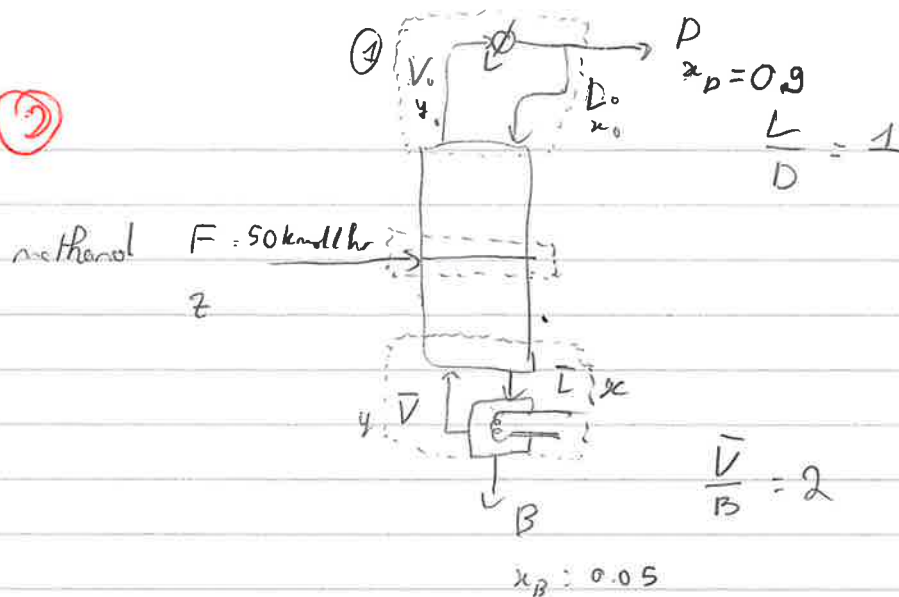


D.6. ②



a)

Top M.B (1):

$$V_0 = L + D$$

$$V_0 y_0 = L x_0 + D x_D$$

CMO condition $\Rightarrow$
$$\begin{cases} V y = L x + D x_D \\ V = L + D \end{cases}$$

$$\Rightarrow y = \frac{L}{V} x + \frac{D}{V} x_D \quad \Rightarrow y = \frac{L}{V} x + \left(1 - \frac{L}{V}\right) x_D$$

$$\frac{D}{V} = \frac{V-L}{V} = 1 - \frac{L}{V}$$

$$\frac{L}{D} = 1 \Rightarrow \frac{D}{L} = 1 \Rightarrow \frac{V-L}{L} = 1 \Rightarrow \frac{V}{L} - 1 = 1 \Rightarrow \frac{V}{L} = 2 \Rightarrow \frac{L}{V} = \frac{1}{2}$$

$\Rightarrow$ Top op. eqn: $y = \frac{1}{2}x + \frac{1}{2}(0.9) \Rightarrow y = \frac{1}{2}x + 0.45$

Bottom M.B (2) under CMO condition

$$\bar{L} = B + \bar{V}$$

$$\bar{L} x = B x_B + \bar{V} y$$

$$\Rightarrow y = \frac{\bar{L}}{\bar{V}} x - \frac{B}{\bar{V}} x_B \Rightarrow y = \frac{\bar{L}}{\bar{V}} x - \left(\frac{\bar{L}-\bar{V}}{\bar{V}}\right) x_B$$

$\Rightarrow$ Bottom op. eqn: $y = \frac{\bar{L}}{\bar{V}} x + \left(1 - \frac{\bar{L}}{\bar{V}}\right) x_B$

$$\frac{\bar{V}}{B} = 2 \Rightarrow \frac{B}{\bar{V}} = \frac{1}{2} \Rightarrow \frac{\bar{L}-\bar{V}}{\bar{V}} = \frac{1}{2} \Rightarrow \frac{\bar{L}}{\bar{V}} - 1 = \frac{1}{2} \Rightarrow \frac{\bar{L}}{\bar{V}} = \frac{3}{2}$$

$$\Rightarrow y = \frac{3}{2}x + (1 - \frac{3}{2})0.05 \Rightarrow y = \frac{3}{2}x - 0.025$$

Overall M.B:

$$F = B + D$$

$$Fz = Bx_B + Dx_D$$

At feed line: top op. eqn and bot. op. eqn intersect

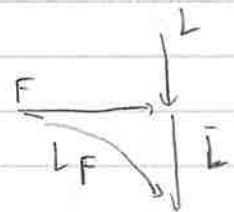
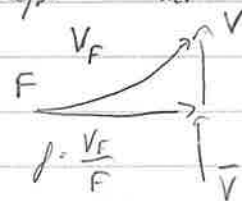
$$\begin{cases} Vy = Lx + Dx_D \\ \bar{V}y = \bar{L}x - Bx_B \end{cases}$$

$$\therefore (V - \bar{V})y = (L - \bar{L})x + Dx_D + Bx_B$$

$$(V - \bar{V})y = (L - \bar{L})x + Fz$$

$$y = \frac{L - \bar{L}}{V - \bar{V}}x + \frac{Fz}{V - \bar{V}}$$

$$y = -\frac{L_F}{V_F}x + \frac{F}{V_F}z = -\frac{q}{f}x + \frac{z}{f} \Rightarrow y = -\frac{q}{1-q}x + \frac{z}{1-q}$$



$$q = \frac{L_F}{F}$$

$$V = V_F + \bar{V}$$

$$\bar{L} = L_F + L$$

Since the feed is saturated liq, $V_F = 0$ & $L_F = 1 \Rightarrow q = 1$

$$\therefore \text{slope} = \frac{-q}{1-q} = -\infty$$

$\Rightarrow$ Feed op. eqn $= x = z$

The top, bot and feed op. eqn all intersect at one point

$$\Rightarrow \frac{3}{2}x - 0.025 = \frac{1}{2}x + 0.45 \Rightarrow x = 0.475 \text{ feed op. eqn.}$$

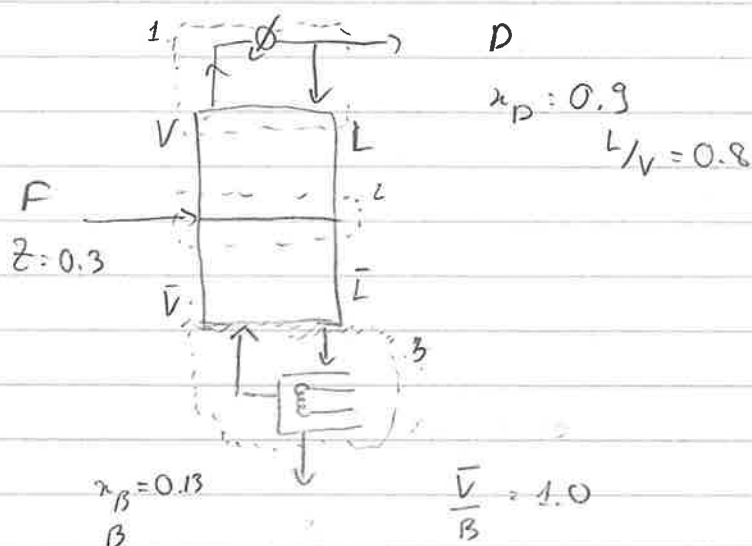
b) From graph, optimum feed stage = 3, total equilibrium stages = 7

$$c) y = \frac{3}{2}x - 0.025 = \frac{1}{2}x + 0.45 \Rightarrow x = 0.475, \text{ since feed line passes thru } y = x = z = 0.475, \boxed{z = 0.475}$$

$$h)(\text{cont}) \frac{x_D - x_S}{x_G - x_S} = \frac{0.5 - 0.1}{0.02 - 0.1} = 0.87 \Rightarrow \text{Can do extra stage}$$

D.7: 7

target component $\rightarrow$ acetone
methanol



a) Top balance eqn (1):

$$\begin{aligned} V &= D + L \\ V y &= D x_D + L x \end{aligned} \quad \Rightarrow \quad y = \frac{L}{V} x + \frac{D}{V} x_D$$

$$y = \frac{L}{V} x + \frac{V-L}{V} x_D = \frac{L}{V} x + \left(1 - \frac{L}{V}\right) x_D$$

$$y = 0.8x + (1 - 0.8) 0.9$$

$$\text{Top op. eqn: } \boxed{y = 0.8x + 0.18}$$

Bottom balance eqn (2):

$$\bar{V} + B = \bar{L}$$

$$\bar{L} x = \bar{V} y + B x_B$$

$$\Rightarrow y = \frac{\bar{L}}{\bar{V}} x - \frac{B}{\bar{V}} x_B$$

$$\frac{\bar{V}}{B} = 1 \Rightarrow \frac{B}{\bar{V}} = 1 \Rightarrow \frac{\bar{L} - \bar{V}}{\bar{V}} = 1 \Rightarrow \frac{\bar{L}}{\bar{V}} - 1 = 1 \Rightarrow \frac{\bar{L}}{\bar{V}} = 2$$

$\Rightarrow$ bottom op. eqn: $y = 2x - 0.13$

Overall M.B:

$$Fz = Dx_D + Bx_B$$

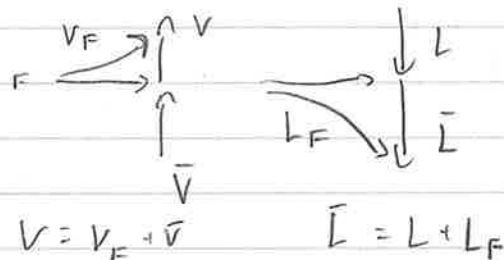
$$F = D + B$$

At feed plate, top & bot op. eqn intersect

$$\begin{cases} Vy = Lx + Dx_D \\ \bar{V}y = \bar{L}x - Bx_B \end{cases}$$

$$\Rightarrow (V - \bar{V})y = (L - \bar{L})x + (Dx_D + Bx_B)$$

$$\Rightarrow y = \frac{L - \bar{L}}{V - \bar{V}} x + \frac{Fz}{V - \bar{V}}$$



intersection of top & bot op. lines

$$2x - 0.13 = 0.8x + 0.18 \Rightarrow x = 0.258 \Rightarrow y = 0.387$$

Feed line passes thru $x=y=z=0.3$ and $(0.258, 0.387)$

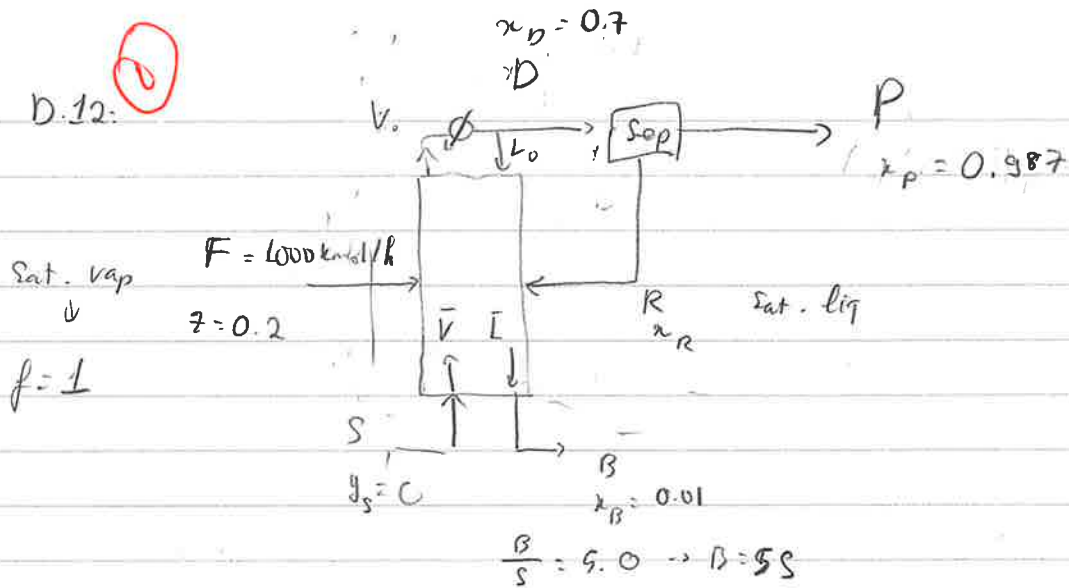
From curve: $x_3 = 0.52$

b) From graph: $y_7 = 0.51$

c) Total stages: 9

Optimal stage: 7

D.12:



CMO condition: $\frac{L_0}{D} = \frac{L}{D} = 1.0$

Top M.B:

$$V = D + L$$

$$V y = L x + D x_D$$

$$y = \frac{L}{V} x + \frac{D}{V} x_D$$

$$\frac{D}{L} = 1 \Rightarrow \frac{V-L}{L} = 1 \Rightarrow \frac{V}{L} - 1 = 1 \Rightarrow \frac{V}{L} = 2 \Rightarrow \frac{L}{V} = \frac{1}{2}$$

$$\frac{D}{V} = \frac{D/L}{V/L} = \frac{1}{2}$$

$$\therefore y = \frac{1}{2} x + \frac{1}{2} (0.7) \Rightarrow y = \frac{1}{2} x + 0.35 \quad \text{Top op. eqn}$$

Bot M.B:

$$\bar{V} + B = S + \bar{L} \quad (S = \bar{V}; \bar{L} = B)$$

$$\bar{V} y + B x_B = \frac{S y_s}{\downarrow 0} + \bar{L} x \Rightarrow y = \frac{\bar{L}}{\bar{V}} x - \frac{B}{\bar{V}} x_B \Rightarrow y = \frac{\bar{L}}{\bar{V}} x - \frac{\bar{L}}{\bar{V}} x_B$$

$$\frac{B}{S} = 5 \Rightarrow \frac{\bar{L}}{\bar{V}} = 5 \Rightarrow y = 5x - 5 \cdot 0.01 \Rightarrow y = 5x - 0.05$$

Overall M.B

$$\begin{cases} F + S = B + P \\ F z + S y_s = B x_B + P x_P \end{cases} \rightarrow \begin{cases} F + \frac{1}{5} B = B + P \\ F z + S y_s = B x_B + P x_P \end{cases} \rightarrow \begin{cases} 1000 = \frac{4}{5} B + P \\ 1000 \cdot 0.2 = 0.01 B + 0.987 P \end{cases}$$

$$\Rightarrow P = 192.4 \text{ kmol/h}$$

$$B = 1009.5 \text{ kmol/h} \rightarrow S = 201.9 \text{ kmol/h}$$

4)

Dis. Column M.B:

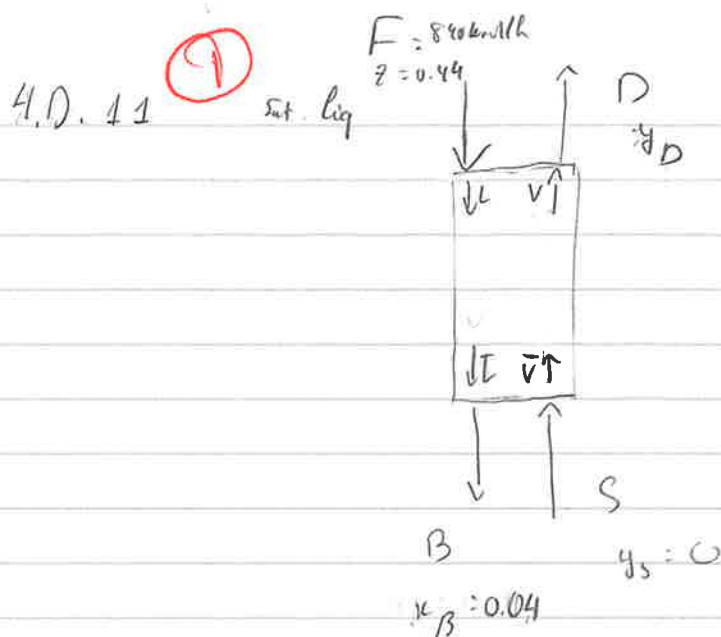
$$\begin{cases} F + R + S = D + B \\ F_z + R_{x_R} + S_{y_S} = D_{x_D} + B_{x_B} \rightarrow F_z + R_{x_R} = D_{x_D} + B_{x_B} \\ \rightarrow F_z + R_{x_R} = (P+R)x_D + B_{x_B} \quad (1) \end{cases}$$

Sep. M.B:

$$\begin{cases} D_{x_D} = P_{x_P} + R_{x_R} \rightarrow (P+R)x_D = P_{x_P} + R_{x_R} \quad (2) \\ D = P + R \end{cases}$$

$$(1), (2) \Rightarrow \begin{cases} 1500 \cdot 0.2 + R_{x_R} = (192.4 + R) \cdot 0.7 + 0.01 \cdot 1009.5 \\ (192.4 + R) \cdot 0.7 = 0.987 \cdot 192.4 + R_{x_R} \end{cases}$$

9



a) CMO condition $\Rightarrow F = L = \bar{L} = B$ & $D = V = \bar{V} = S$

Overall M.B:

$$F + S = B + D \quad \rightarrow \quad F + \frac{2}{3}F = F + D$$

$$Fz + Sy_S = Bx_B + Dy_D \quad Fz = Fx_B + Dy_D$$

$$\Rightarrow \begin{cases} 840 + \frac{2}{3}840 = 840 + D \\ 840 \cdot 0.44 = 840 \cdot 0.04 + D y_D \end{cases} \Rightarrow \begin{cases} D = 560 \text{ kmol/h} \\ y_D = 0.6 \end{cases}$$

Bot. M.B:

$$\bar{L} + S = B + \bar{V}$$

$$\bar{L}x + Sy_S = Bx_B + \bar{V}y \quad \Rightarrow \quad y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}}x_B = \frac{F}{S}x - \frac{F}{S}x_B$$

$$\Rightarrow y = \frac{3}{2}x - \frac{3}{2}0.04 \quad \Rightarrow \quad y = \frac{3}{2}x - 0.06$$

Top (feed) M.B

$$y = y_D \quad (\text{derived in Section 4.9.5 in the book})$$

Bot op & Top op. eqn intersection:

$$y = 0.6 = \frac{3}{2}x - 0.06 \Rightarrow x = 0.44$$

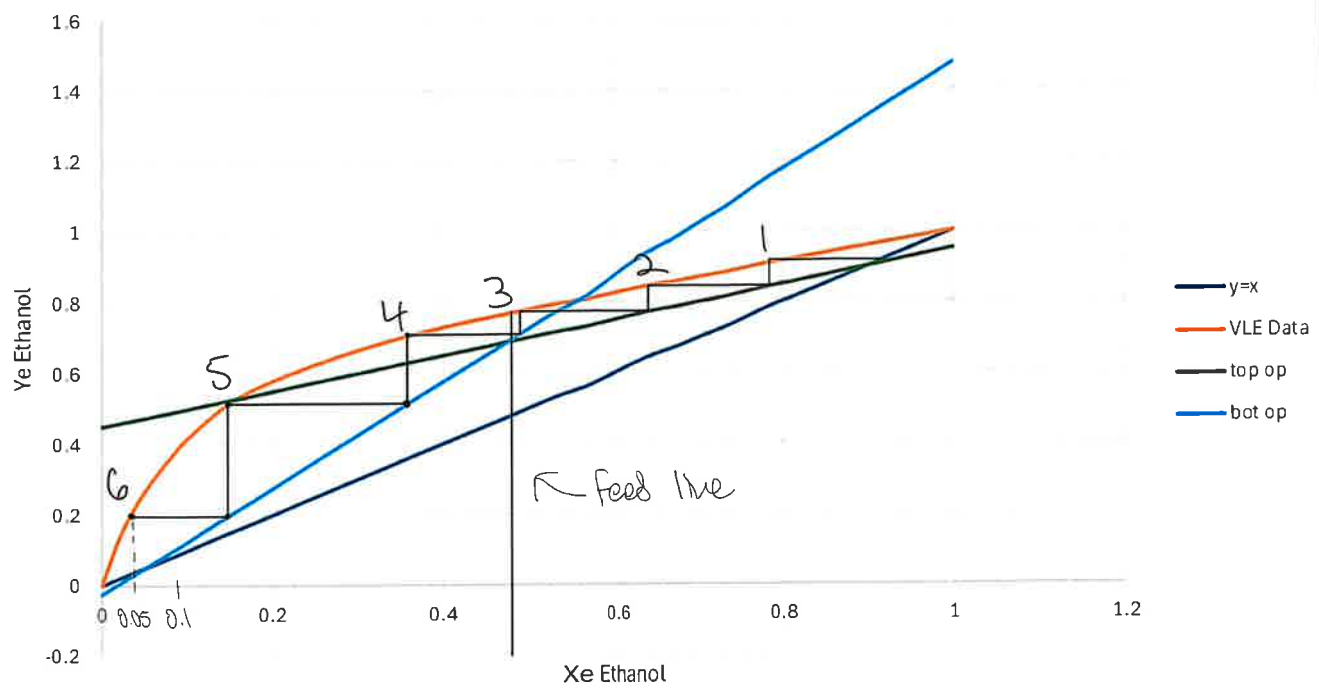
Feed line passes thru $x=y=z: 0.44$ and $(0.44, 0.6)$

$\Rightarrow$ Feed line: $x = 0.44$

b) From graph: 2 stages required

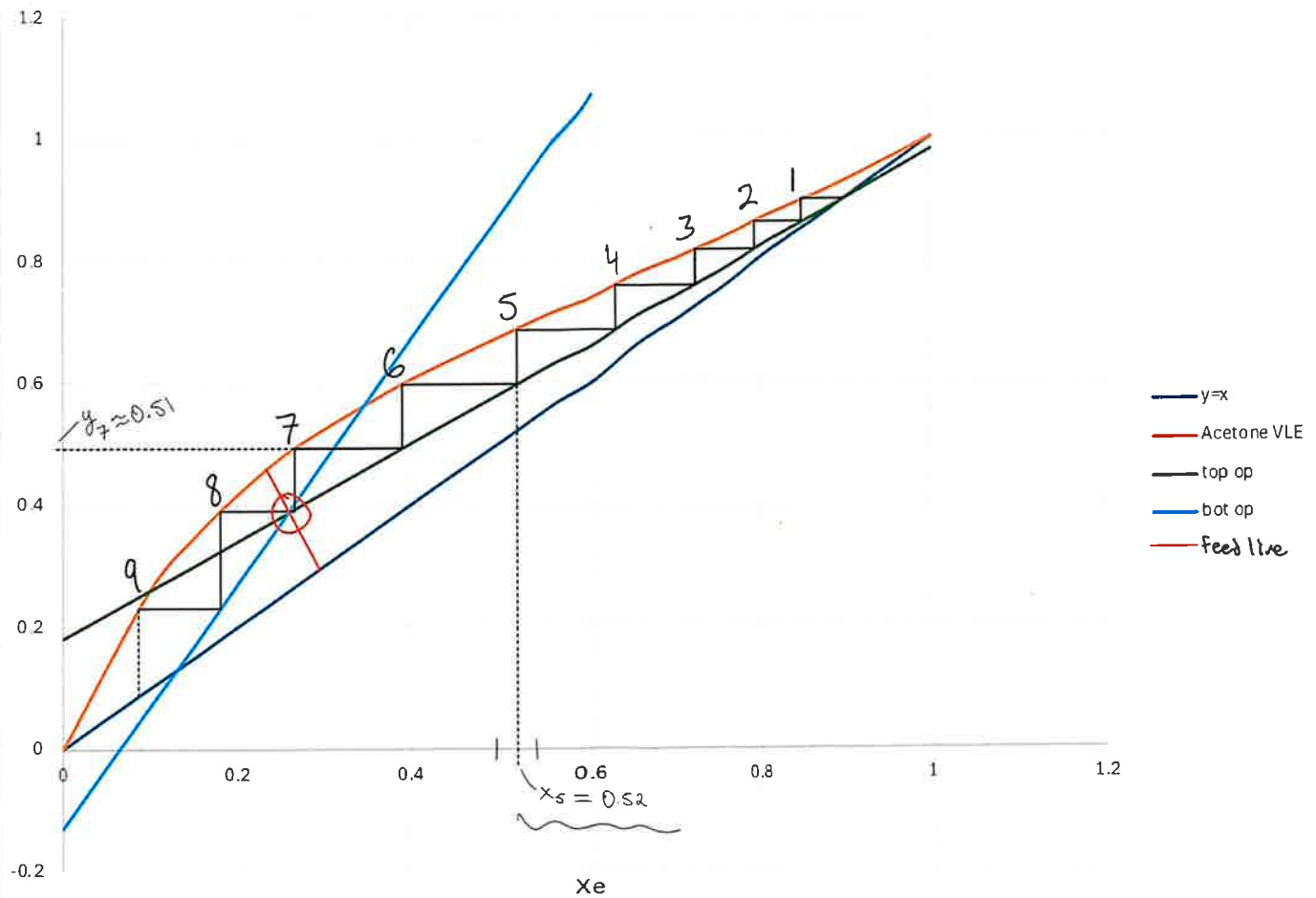
c) Number of equilibrium stages $\rightarrow \infty \Rightarrow$ bot op. line very close to $y=x$ curve
 $x_B \approx 0.32$ ~~not~~ 0.07

4.D.6

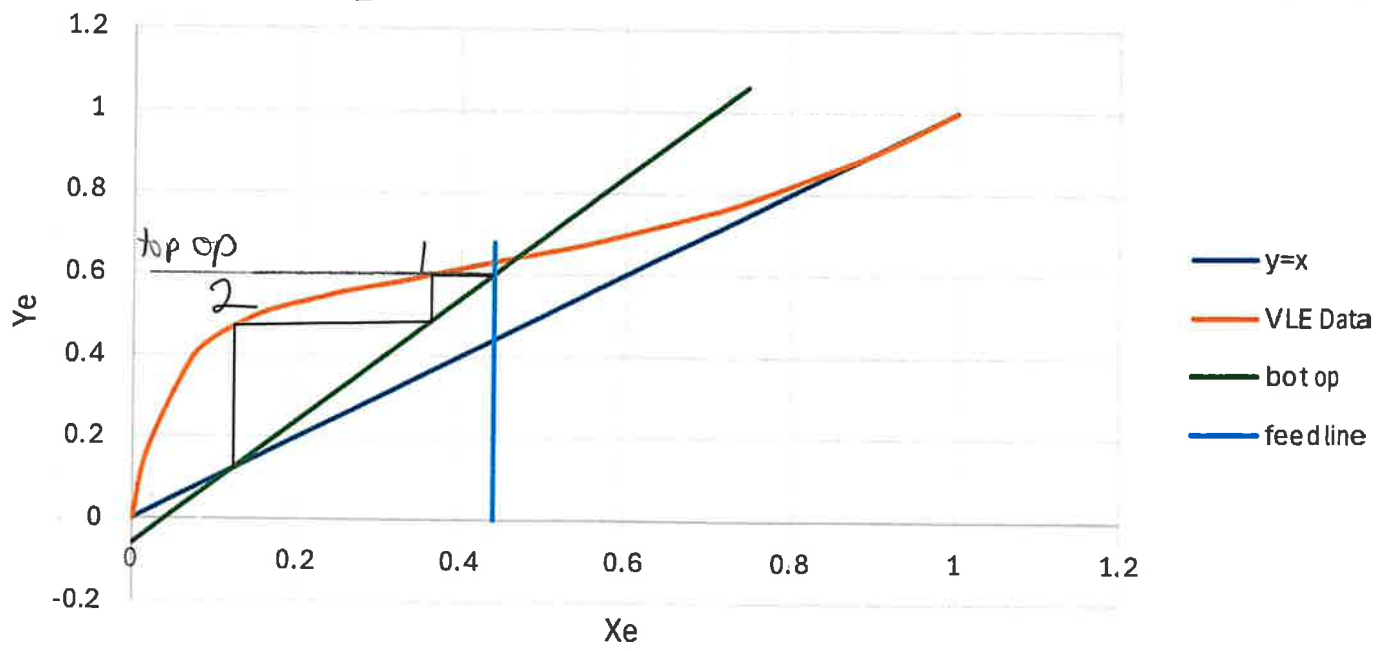


4.D.7

Y_e vs X_e for Acetone (ethanol mix)



4.D.11



12